Total No. of Questions : 8 ] [ Total No. of Printed Pages : 3

# Paper Code : 21327

## F-427

## B.C.A. (Sixth Semester) Examination, 2023

## (New Course)

## Paper No. BCA-N-602

## ARTIFICIAL INTELLIGENCE

*Time : Three Hours ]* *[ Maximum Marks : 70*

**Note :** Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What is 'AI'? Explain its importance in today's scenario with proper examples.

   (b) Differentiate heuristic algorithm and solution guaranteed algorithm.



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2. (a) Describe the parsing techniques.

   (b) Write in brief about :

   (i) Fillmore's Grammers

   (ii) Context free grammer

3. (a) What do you mean by "First order logic"? Write one example.

   (b) What is the difference between forward deduction and backward deduction?

4. (a) Define "PROLOG" with its application. What are miniskey frames?

   (b) How "MYCIN" expert system is different from "DENDRAL" expert system?

5. (a) Compare "NLP' with 'expert' system giving their applications.

   (b) Write down syntax of both 'LISP' and 'PROLOG'. Explain the applications of LISP in programming world.



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6. (a) How pattern-recognition is implemented using the concept of artificial intelligence?

(b) Compare various techniques used for knowledge representation.

7. (a) Explain transition nots and augmented transitions nots with suitable examples.

(b) Illustrate the "Horn-Clause" with two examples.

8. Write short notes on : (any **four**)

(i) Meta knowledge

(ii) Applications of AI in education

(iii) Speech recognition

(iv) Shanks conceptual dependency

(v) CFG



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## F-427

## B. C. A. (Sixth Semester)

## Examination, 2022

## (New Course)

**Paper No. BCA-N-602**

**ARTIFICIAL INTELLIGENCE**

***Time : Three Hours ]*** ***[ Maximum Marks : 70***

**Note** : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meaning.

1. (a) What do you mean by the term 'AI'? How it is useful in today's world?

   (b) Differentiate heuristic algorithm from solution guaranteed algorithm.



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2. (a) Describe natural language processing with suitable example.

   (b) Define "Parsing". What are various parsing techniques?

3. (a) Explain the context free grammer (CFG) with proper example.

   (b) How Fillmore's grammer is different from CFG?

4. (a) Explain the first order predicate calculus with one example.

   (b) What are the features of " PROLOG"?

5. (a) Describe the "Minskey Frames" with examples. https://www.mjpruonline.com

   (b) How to represent knowledge using Horn-clauses?

6. (a) What are "Expert System"? Elaborate "MYCIN" with suitable example.



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(b) How "Meta-knowledge" is useful in any Expert System.

7. (a) Explain the term "Pattern Recognition" in detail with its principle using one example.

(b) How "Lisp" is different from "PROLOG"? Explain clearly with their syntax.

8. Write in brief about any **four** of the following :

(a) Shanks Conceptual Dependency

(b) Semantic Nets

(c) DENDRAL

(d) Speech Recognition

(e) Machine Learning



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Total No. of Questions : 8] [ Total No. of Printed Pages : 4

# Paper Code : 21327

## F-427

## B.C.A. (Sixth Semester) Examination-2021

**(New Course)**

**Paper No. BCA-N-602**

**ARTIFICIAL INTELLIGENCE**

*Time : 1½ Hours ]* *[ Maximum Marks : 70*

**Note :** Attempt **all** sections as directed.

### Section-A

**Note :** Attempt any **one** question. 20 each

1. (a) How the concept of Artificial Intelligence is applied in different domains? Justify with suitable examples.



(b) Differentiate solution guaranteed algorithms from heuristic algorithms with proper case study.

2. (a) What is automated reasoning?

(b) Explain different types of Parsing-techniques.

3. (a) Elaborate the terms:-

(i) Context free grammar

(ii) Fillmore's grammars

(b) What are grammar free analyzers?

### Section-B

**Note :** Attempt any **one** question. 20 each

4. (a) Describe the "natural language processing concept in detail.

(b) What do you understand by "first or-



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der predicate logic"? Describe the term quantifiers with its types.

5. (a) Explain the "Horn Clauses" with example.

(b) How PROLOG is useful in knowledge representation?

6. (a) Explain semantic nets and partitioned nets with example.

(b) What are expert systems? How these are useful in medical Science?

## Section-C

**Note :** Attempt any **one** question. 30 each

7. (a) Write in detail the theory behind the "MYCIN".

(b) What is "Meta knowledge"? How it is significant in any expert system.



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8. (a) How any pattern recognition system works ? Explain with an example.

(b) Write short notes on :

(i) LISP

(ii) Shanks Conceptual dependency

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# Paper Code : 21327

## F–427

## B. C. A. (Sixth Semester) EXAMINATION, 2019

### (New Course)

Paper No. BCA—N—602

ARTIFICIAL INTELLIGENCE

*Time : Three Hours* ] [ *Maximum Marks : 70*

**Note :** Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) What is natural language ? What are the general approaches to natural language understanding ?

   (b) Explain Heuristic algorithm *vs.* solution guaranteed algorithm.

2. (a) Explain the following :

   (i) Automated reasoning

   (ii) Visual perception

   (b) What are the various approaches used for knowledge representation ?



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3. (a) Why LISP is better than C while doing programming in the field of AI ?

   (b) Explain augmented transition nets.

4. (a) Explain the various properties of attributes which are independent of specific knowledge they encode.

   (b) Explain the following :

   (i) Parsing techniques

   (ii) Transformational Grammar

5. (a) Explain Production rules knowledge base.

   (b) Explain Minskey frames with example

6. (a) What is expert system ? Explain with example.

   (b) Explain the existing system DENDRAL

7. (a) What is pattern recognition ? Explain speech recognition and objection identification.

   (b) What is PROLOG ? Explain its utility in pattern recognition. http://www.mjpruonline.com

8. Write short notes on any *two* of the following :

   (i) Interception semantic and model

   (ii) Self Explaining System

   (iii) First order predicate calculus

   (iv) MYCIN Expert System.

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# Paper Code : 21327

## F–427

## B. C. A. (Sixth Semester) EXAMINATION, 2018

### (New Course)

Paper No. BCA—N—602

ARTIFICIAL INTELLIGENCE

*Time : Three Hours* ] [ *Maximum Marks : 70*

**Note :** Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meaning.

1. (a) Define artificial intelligence. Explain various characteristics of an expert system.

   (b) What is heuristic algorithm ? Enumerate various characteristics of heuristic algorithm.

2. (a) What is visual perception ? Explain its role in artificial intelligence.

   (b) What is automated reasoning ?

   (c) How can one apply intelligent behaviour of an expert system in problem solving ?

3. (a) Explain role of natural languages in expert system development.

   (b) Explain rules for Fillmore's grammar.



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4. (a) Write a program in PROLOG to add two numbers.

   (b) What are various first order predicate calculus ruler ? Enumerate and explain.

5. (a) Explain Minsky frames.

   (b) What are various rules for forward and backward deductions ? Explain.

6. (a) What do you mean by self explaining system ? Explain.

   (b) Explain the following expert system :

   (i) DENDRAL

   (ii) MYCIN

7. (a) What is pattern recognition ? Explain.

   (b) Explain the following applications of Expert system :

   (i) Speech Recognition

   (ii) Symbolic Description

8. Write short notes on any *four* of the following :

   (i) LISP

   (ii) Semantic Model

   (iii) Horn Clauses

   (iv) Grammar free analyzers

   (v) Minskey frames

   (vi) Self-explaining system



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# Paper Code : 21327

## F–427

## B. C. A. (Sixth Semester)
## - EXAMINATION, 2016

### (New Course)

Paper No. BCA–N–602

ARTIFICIAL INTELLIGENCE

*Time : Three Hours* ] [ *Maximum Marks : 70*

**Note :** Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) What is the concept of Artificial intelligence ? What are the most important areas where Artificial intelligence is applied ?

   (b) Define Expert system. Explain the operation of an Expert system with the help of block diagram. Also explain MYCIN expert system.

2. (a) Indentify and describe two good application areas for expert system within a university environment.



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   (b) What is the difference between declarative and procedural knowledge ? Briefly describe the meaning of knowledge representation.

3. (a) What are different advanced programming techniques that are used in PROLOG ? Also discuss control strategies of PROLOG.

   (b) What do you understand by Natural Languages ? Also explain context free and transformational grammars ?

4. (a) What do you understand by the term pattern Recognition ? Also explain the structured description and symbolic description.

   (b) What is heuristic search ? Explain the problem characteristics of heuristic search.

5. (a) What do you understand by the term production rule knowledge base and Forward and Backward Deduction ? Explain.

   (b) What is predicate logic ? Explain resolution in predicate logic.

6. (a) What do you understand by terms transition nets and augmented transitions nets ? Explain their differences ? http://www.mjpruonline.com

   (b) What are the problems in Natural language understanding ? What are their applications ? Also explain automated reasoning ?

7. (a) What is grammar free analyzers ? Also explain sentence generation and translation ?

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(b) What is LISP ? What are the advantages of LISP ? How LISP functions are used in Artificial intelligence ?

8. Write short notes on the following :
   (i) Case Grammar theory
   (ii) Semantic Nets
   (iii) Expertise Transfer
   (iv) Fillmore's grammars

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